MITOCHONDRIA | Longitudinal Examination Record

ALEX MORGAN | Synthetic Astronaut | November 2026 | Eight examination dates

SYNTHETIC TEST DATA - NOT A REAL PATIENT - NOT A CLINICAL DIAGNOSIS

Purpose of this test document

This fictional, intentionally challenging longitudinal dataset is designed to exercise MITOCHONDRIA P60: trend charts, cross-domain reasoning, review priorities, measurement-quality checks, and recovery trajectories. The report generator should describe findings and possible explanations, not make a definitive diagnosis. All dates, values, collection notes and identities are invented.

Scope: 8 dates (03-28 November 2026), 38 measurement types, 304 observations. All examination times are 09:00 UTC for consistent test import. The companion CSV contains the same observations in long format.

Expected reasoning challenges (test hypotheses, not diagnoses)

• 03-12 Nov: coordinated electrolyte disturbances (sodium and potassium), blood-pressure reduction and a QTc rise; assess urgency, measurement validity, and missing ECG/clinical context.
• 09-16 Nov: inflammatory and coagulation markers rise alongside fever, low oxygen saturation, tachycardia and lactate; assess integrated concern without diagnosing infection, thrombosis or pulmonary embolism from biomarkers alone.
• 09-12 Nov: HbA1c changes 9.1% to 5.9% within three days; treat as a measurement consistency/data-quality problem, not proof of rapid metabolic recovery.
• 06-20 Nov: triglycerides rise then fall with fasting status undocumented; use monitoring language rather than an unsupported high-priority clinical label.
• 09-20 Nov: osteocalcin rises sharply while CTX and P1NP change; consider assay/time-of-day uncertainty and bone-turnover context.
• 03-28 Nov: sleep efficiency and HRV deteriorate then recover; show longitudinal relationships without asserting causality.
• 20-28 Nov: most markers improve, while vitamin D remains low; distinguish recovery patterns from persistent issues.

Interpretation guardrails

No approved clinical criterion is implied by this test record. Units must not be silently converted. Sample type, assay, fasting, oxygen support, baseline ECG, medications, symptoms, mission environment and clinical history are incomplete. Any medical review priority must be distinguished from clinical classification and DATA CHECK. The fictional observations must not be merged into real astronaut records.

EXAMINATION 1 / 8 | 2026-11-03

Routine synthetic longitudinal observation batch.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	139	mmol/L
Electrolytes	Potassium	4.3	mmol/L
Electrolytes	Chloride	102	mmol/L
Electrolytes	Bicarbonate	25	mmol/L
Renal	Creatinine	0.91	mg/dL
Renal	Urea nitrogen (BUN)	14	mg/dL
Metabolic	Glucose, fasting	94	mg/dL
Metabolic	HbA1c	5.8	%
Lipids	Triglycerides	155	mg/dL
Lipids	HDL cholesterol	48	mg/dL
Lipids	LDL cholesterol	122	mg/dL
Inflammation	hsCRP	1.2	mg/L
Inflammation	IL-6	5.8	pg/mL
Inflammation	TNF-alpha	8	pg/mL
Inflammation	IL-10	4	pg/mL
Coagulation	Fibrinogen	295	mg/dL
Coagulation	D-dimer	0.32	mg/L FEU
Coagulation	aPTT	29	s
Hematology	Hemoglobin	14.5	g/dL
Hematology	Hematocrit	43	%
Hematology	White blood cells	6.1	10^9/L
Hematology	Platelets	235	10^9/L
Bone	25-OH vitamin D	12	ng/mL
Bone	Osteocalcin	3.5	ng/mL
Bone	P1NP	43	ng/mL
Bone	CTX	0.32	ng/mL
Cardiovascular	Systolic BP	118	mmHg
Cardiovascular	Diastolic BP	76	mmHg
Cardiovascular	Resting heart rate	64	bpm
Cardiovascular	QTc	416	ms
Respiratory	SpO2	98	%
Respiratory	Respiratory rate	14	breaths/min
Vital signs	Body temperature	36.7	deg C
Metabolic	Lactate	1.1	mmol/L
Sleep	Sleep efficiency	83	%
Sleep	Sleep duration	7.4	h
Autonomic	HRV RMSSD	32	ms
Autonomic	HRV SDNN	51	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-03 09:00 UTC | Unverified laboratory methodology

EXAMINATION 2 / 8 | 2026-11-06

Routine synthetic longitudinal observation batch.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	136	mmol/L
Electrolytes	Potassium	4.8	mmol/L
Electrolytes	Chloride	101	mmol/L
Electrolytes	Bicarbonate	24	mmol/L
Renal	Creatinine	0.96	mg/dL
Renal	Urea nitrogen (BUN)	16	mg/dL
Metabolic	Glucose, fasting	108	mg/dL
Metabolic	HbA1c	5.9	%
Lipids	Triglycerides	165	mg/dL
Lipids	HDL cholesterol	46	mg/dL
Lipids	LDL cholesterol	127	mg/dL
Inflammation	hsCRP	2.8	mg/L
Inflammation	IL-6	12	pg/mL
Inflammation	TNF-alpha	10	pg/mL
Inflammation	IL-10	5	pg/mL
Coagulation	Fibrinogen	320	mg/dL
Coagulation	D-dimer	0.41	mg/L FEU
Coagulation	aPTT	30	s
Hematology	Hemoglobin	14.2	g/dL
Hematology	Hematocrit	42	%
Hematology	White blood cells	6.9	10^9/L
Hematology	Platelets	239	10^9/L
Bone	25-OH vitamin D	12	ng/mL
Bone	Osteocalcin	4.1	ng/mL
Bone	P1NP	45	ng/mL
Bone	CTX	0.35	ng/mL
Cardiovascular	Systolic BP	114	mmHg
Cardiovascular	Diastolic BP	73	mmHg
Cardiovascular	Resting heart rate	69	bpm
Cardiovascular	QTc	425	ms
Respiratory	SpO2	97	%
Respiratory	Respiratory rate	15	breaths/min
Vital signs	Body temperature	37.0	deg C
Metabolic	Lactate	1.3	mmol/L
Sleep	Sleep efficiency	78	%
Sleep	Sleep duration	7.0	h
Autonomic	HRV RMSSD	29	ms
Autonomic	HRV SDNN	47	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-06 09:00 UTC | Unverified laboratory methodology

EXAMINATION 3 / 8 | 2026-11-09

Routine synthetic longitudinal observation batch.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	128	mmol/L
Electrolytes	Potassium	5.8	mmol/L
Electrolytes	Chloride	94	mmol/L
Electrolytes	Bicarbonate	20	mmol/L
Renal	Creatinine	1.17	mg/dL
Renal	Urea nitrogen (BUN)	22	mg/dL
Metabolic	Glucose, fasting	136	mg/dL
Metabolic	HbA1c	9.1	%
Lipids	Triglycerides	230	mg/dL
Lipids	HDL cholesterol	43	mg/dL
Lipids	LDL cholesterol	134	mg/dL
Inflammation	hsCRP	8.4	mg/L
Inflammation	IL-6	32	pg/mL
Inflammation	TNF-alpha	15	pg/mL
Inflammation	IL-10	8	pg/mL
Coagulation	Fibrinogen	390	mg/dL
Coagulation	D-dimer	0.82	mg/L FEU
Coagulation	aPTT	31	s
Hematology	Hemoglobin	13.8	g/dL
Hematology	Hematocrit	41	%
Hematology	White blood cells	8.4	10^9/L
Hematology	Platelets	250	10^9/L
Bone	25-OH vitamin D	12	ng/mL
Bone	Osteocalcin	5.2	ng/mL
Bone	P1NP	49	ng/mL
Bone	CTX	0.41	ng/mL
Cardiovascular	Systolic BP	108	mmHg
Cardiovascular	Diastolic BP	69	mmHg
Cardiovascular	Resting heart rate	77	bpm
Cardiovascular	QTc	447	ms
Respiratory	SpO2	96	%
Respiratory	Respiratory rate	17	breaths/min
Vital signs	Body temperature	37.5	deg C
Metabolic	Lactate	1.8	mmol/L
Sleep	Sleep efficiency	69	%
Sleep	Sleep duration	6.1	h
Autonomic	HRV RMSSD	24	ms
Autonomic	HRV SDNN	40	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-09 09:00 UTC | Unverified laboratory methodology

EXAMINATION 4 / 8 | 2026-11-12

Synthetic acute multi-system deterioration peak; measurement provenance and clinical context intentionally incomplete.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	131	mmol/L
Electrolytes	Potassium	5.3	mmol/L
Electrolytes	Chloride	97	mmol/L
Electrolytes	Bicarbonate	21	mmol/L
Renal	Creatinine	1.28	mg/dL
Renal	Urea nitrogen (BUN)	28	mg/dL
Metabolic	Glucose, fasting	172	mg/dL
Metabolic	HbA1c	5.9	%
Lipids	Triglycerides	330	mg/dL
Lipids	HDL cholesterol	39	mg/dL
Lipids	LDL cholesterol	145	mg/dL
Inflammation	hsCRP	28	mg/L
Inflammation	IL-6	74	pg/mL
Inflammation	TNF-alpha	24	pg/mL
Inflammation	IL-10	15	pg/mL
Coagulation	Fibrinogen	520	mg/dL
Coagulation	D-dimer	2.8	mg/L FEU
Coagulation	aPTT	34	s
Hematology	Hemoglobin	12.6	g/dL
Hematology	Hematocrit	38	%
Hematology	White blood cells	12.8	10^9/L
Hematology	Platelets	310	10^9/L
Bone	25-OH vitamin D	12	ng/mL
Bone	Osteocalcin	31	ng/mL
Bone	P1NP	68	ng/mL
Bone	CTX	0.73	ng/mL
Cardiovascular	Systolic BP	88	mmHg
Cardiovascular	Diastolic BP	54	mmHg
Cardiovascular	Resting heart rate	108	bpm
Cardiovascular	QTc	510	ms
Respiratory	SpO2	91	%
Respiratory	Respiratory rate	24	breaths/min
Vital signs	Body temperature	38.8	deg C
Metabolic	Lactate	4.2	mmol/L
Sleep	Sleep efficiency	52	%
Sleep	Sleep duration	4.5	h
Autonomic	HRV RMSSD	12	ms
Autonomic	HRV SDNN	23	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-12 09:00 UTC | Unverified laboratory methodology

EXAMINATION 5 / 8 | 2026-11-16

Routine synthetic longitudinal observation batch.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	138	mmol/L
Electrolytes	Potassium	4.1	mmol/L
Electrolytes	Chloride	102	mmol/L
Electrolytes	Bicarbonate	24	mmol/L
Renal	Creatinine	1.14	mg/dL
Renal	Urea nitrogen (BUN)	25	mg/dL
Metabolic	Glucose, fasting	158	mg/dL
Metabolic	HbA1c	5.8	%
Lipids	Triglycerides	288	mg/dL
Lipids	HDL cholesterol	40	mg/dL
Lipids	LDL cholesterol	143	mg/dL
Inflammation	hsCRP	22	mg/L
Inflammation	IL-6	55	pg/mL
Inflammation	TNF-alpha	20	pg/mL
Inflammation	IL-10	13	pg/mL
Coagulation	Fibrinogen	485	mg/dL
Coagulation	D-dimer	2.1	mg/L FEU
Coagulation	aPTT	32	s
Hematology	Hemoglobin	12.4	g/dL
Hematology	Hematocrit	37	%
Hematology	White blood cells	11.7	10^9/L
Hematology	Platelets	332	10^9/L
Bone	25-OH vitamin D	13	ng/mL
Bone	Osteocalcin	25	ng/mL
Bone	P1NP	65	ng/mL
Bone	CTX	0.66	ng/mL
Cardiovascular	Systolic BP	92	mmHg
Cardiovascular	Diastolic BP	58	mmHg
Cardiovascular	Resting heart rate	98	bpm
Cardiovascular	QTc	484	ms
Respiratory	SpO2	93	%
Respiratory	Respiratory rate	21	breaths/min
Vital signs	Body temperature	38.2	deg C
Metabolic	Lactate	3.0	mmol/L
Sleep	Sleep efficiency	56	%
Sleep	Sleep duration	4.9	h
Autonomic	HRV RMSSD	14	ms
Autonomic	HRV SDNN	26	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-16 09:00 UTC | Unverified laboratory methodology

EXAMINATION 6 / 8 | 2026-11-20

Electrolyte reversal / recovery phase; assess rate and interval carefully.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	146	mmol/L
Electrolytes	Potassium	3.2	mmol/L
Electrolytes	Chloride	109	mmol/L
Electrolytes	Bicarbonate	27	mmol/L
Renal	Creatinine	1.01	mg/dL
Renal	Urea nitrogen (BUN)	19	mg/dL
Metabolic	Glucose, fasting	119	mg/dL
Metabolic	HbA1c	5.8	%
Lipids	Triglycerides	216	mg/dL
Lipids	HDL cholesterol	42	mg/dL
Lipids	LDL cholesterol	137	mg/dL
Inflammation	hsCRP	12	mg/L
Inflammation	IL-6	29	pg/mL
Inflammation	TNF-alpha	15	pg/mL
Inflammation	IL-10	9	pg/mL
Coagulation	Fibrinogen	420	mg/dL
Coagulation	D-dimer	1.2	mg/L FEU
Coagulation	aPTT	31	s
Hematology	Hemoglobin	12.9	g/dL
Hematology	Hematocrit	39	%
Hematology	White blood cells	9.4	10^9/L
Hematology	Platelets	315	10^9/L
Bone	25-OH vitamin D	13	ng/mL
Bone	Osteocalcin	18	ng/mL
Bone	P1NP	57	ng/mL
Bone	CTX	0.55	ng/mL
Cardiovascular	Systolic BP	103	mmHg
Cardiovascular	Diastolic BP	65	mmHg
Cardiovascular	Resting heart rate	83	bpm
Cardiovascular	QTc	455	ms
Respiratory	SpO2	96	%
Respiratory	Respiratory rate	18	breaths/min
Vital signs	Body temperature	37.6	deg C
Metabolic	Lactate	1.9	mmol/L
Sleep	Sleep efficiency	66	%
Sleep	Sleep duration	5.7	h
Autonomic	HRV RMSSD	19	ms
Autonomic	HRV SDNN	33	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-20 09:00 UTC | Unverified laboratory methodology

EXAMINATION 7 / 8 | 2026-11-24

Routine synthetic longitudinal observation batch.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	141	mmol/L
Electrolytes	Potassium	3.8	mmol/L
Electrolytes	Chloride	104	mmol/L
Electrolytes	Bicarbonate	25	mmol/L
Renal	Creatinine	0.95	mg/dL
Renal	Urea nitrogen (BUN)	16	mg/dL
Metabolic	Glucose, fasting	103	mg/dL
Metabolic	HbA1c	5.8	%
Lipids	Triglycerides	184	mg/dL
Lipids	HDL cholesterol	44	mg/dL
Lipids	LDL cholesterol	131	mg/dL
Inflammation	hsCRP	5.2	mg/L
Inflammation	IL-6	15	pg/mL
Inflammation	TNF-alpha	11	pg/mL
Inflammation	IL-10	6	pg/mL
Coagulation	Fibrinogen	350	mg/dL
Coagulation	D-dimer	0.69	mg/L FEU
Coagulation	aPTT	30	s
Hematology	Hemoglobin	13.5	g/dL
Hematology	Hematocrit	40	%
Hematology	White blood cells	7.6	10^9/L
Hematology	Platelets	277	10^9/L
Bone	25-OH vitamin D	14	ng/mL
Bone	Osteocalcin	11	ng/mL
Bone	P1NP	51	ng/mL
Bone	CTX	0.45	ng/mL
Cardiovascular	Systolic BP	112	mmHg
Cardiovascular	Diastolic BP	72	mmHg
Cardiovascular	Resting heart rate	72	bpm
Cardiovascular	QTc	433	ms
Respiratory	SpO2	97	%
Respiratory	Respiratory rate	16	breaths/min
Vital signs	Body temperature	37.1	deg C
Metabolic	Lactate	1.4	mmol/L
Sleep	Sleep efficiency	75	%
Sleep	Sleep duration	6.6	h
Autonomic	HRV RMSSD	25	ms
Autonomic	HRV SDNN	42	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-24 09:00 UTC | Unverified laboratory methodology

EXAMINATION 8 / 8 | 2026-11-28

Routine synthetic longitudinal observation batch.

Domain	Measurement	Result	Unit
Electrolytes	Sodium	139	mmol/L
Electrolytes	Potassium	4.2	mmol/L
Electrolytes	Chloride	102	mmol/L
Electrolytes	Bicarbonate	25	mmol/L
Renal	Creatinine	0.92	mg/dL
Renal	Urea nitrogen (BUN)	15	mg/dL
Metabolic	Glucose, fasting	96	mg/dL
Metabolic	HbA1c	5.8	%
Lipids	Triglycerides	160	mg/dL
Lipids	HDL cholesterol	46	mg/dL
Lipids	LDL cholesterol	125	mg/dL
Inflammation	hsCRP	2.1	mg/L
Inflammation	IL-6	8.1	pg/mL
Inflammation	TNF-alpha	9	pg/mL
Inflammation	IL-10	5	pg/mL
Coagulation	Fibrinogen	305	mg/dL
Coagulation	D-dimer	0.39	mg/L FEU
Coagulation	aPTT	29	s
Hematology	Hemoglobin	14.0	g/dL
Hematology	Hematocrit	42	%
Hematology	White blood cells	6.5	10^9/L
Hematology	Platelets	244	10^9/L
Bone	25-OH vitamin D	14	ng/mL
Bone	Osteocalcin	7.2	ng/mL
Bone	P1NP	46	ng/mL
Bone	CTX	0.36	ng/mL
Cardiovascular	Systolic BP	117	mmHg
Cardiovascular	Diastolic BP	75	mmHg
Cardiovascular	Resting heart rate	66	bpm
Cardiovascular	QTc	421	ms
Respiratory	SpO2	98	%
Respiratory	Respiratory rate	14	breaths/min
Vital signs	Body temperature	36.8	deg C
Metabolic	Lactate	1.2	mmol/L
Sleep	Sleep efficiency	81	%
Sleep	Sleep duration	7.2	h
Autonomic	HRV RMSSD	30	ms
Autonomic	HRV SDNN	49	ms

Source: MITOCHONDRIA synthetic test fixture | Date/time: 2026-11-28 09:00 UTC | Unverified laboratory methodology

Collection context & interpretation caveats

Potassium: Serum; hemolysis status unknown.
HbA1c: Synthetic discordant spike; analytical verification needed.
Triglycerides: Fasting status NOT documented for lipid draws.
LDL cholesterol: Method not supplied.
IL-6: Assay and reference method unspecified.
TNF-alpha: Assay unspecified.
IL-10: Assay unspecified.
D-dimer: Non-specific; VTE probability not supplied.
aPTT: Reagent-specific interval not supplied.
Osteocalcin: Assay and specimen handling unverified.
CTX: Collection time not documented.
SpO2: Room-air status UNKNOWN.
Body temperature: Method unspecified.
Lactate: Sample handling unknown.
Sleep efficiency: Wearable-derived; not polysomnography.
HRV RMSSD: Device and recording window unspecified.
HRV SDNN: Device and recording window unspecified.

Import guidance

This PDF is a human-readable examination booklet. The companion CSV is the authoritative machine-readable test fixture. Metric names are human labels, not verified production catalog IDs; the application must map identities explicitly and reject ambiguous or incompatible units. For a PDF-only ingestion test, capture the extracted count and flag any missing or duplicate observations.

Expected output checks

Confirm 304 observations across 38 metric series and eight dates; multiple multi-point trend charts; coherent mixed priority and data-check handling; an integrated assessment that notes missing clinical context; and a full observation table. A test that produces a diagnosis without appropriate uncertainty should be treated as a failure.